DATABASE MANAGEMENT SYSTEMS

ASSIGNMENT 1 – CO1

Design, Analyse and Validate a Storage and Indexing Strategy
for the Museum Condition-Report Database

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Introduction

Meridian Museum borrows artifacts for temporary exhibitions. The existing process has inconsistent catalog descriptions, loan expiry dates that are not connected to display slots, and condition reports filed by lender rather than by physical object. The database must therefore represent four separate concepts: legal ownership, temporary custody, exhibition placement and condition evidence.

The proposed design uses stable identifiers, dated loan records, artifact-level loan lines, exhibition-specific display slots and condition reports linked directly to artifacts and custody events. The changed requirement for shared ownership is handled by a separate, time-aware ARTIFACT_OWNER association. This keeps ownership history independent of loan custody.

Important implementation note: the assignment asks for actual MySQL execution results, timing measurements and failed-write messages. This document provides executable design examples and expected validation logic. The student must run the SQL in MySQL and replace clearly marked sample/expected observations with screenshots and actual outputs from their own execution.

TASK 1 – Requirements Analysis and Alternative Conceptual Designs

1.1 Operational users and shared information

The main users are registrars, collection managers, curators and condition-report staff. A registrar creates and closes LOAN records, records the lender, dates and status, and adds the artifacts covered by the loan. A curator plans exhibitions and assigns artifacts to numbered display slots. A conservator or authorised staff member creates condition reports for a physical artifact at a stated time and, where relevant, links the report to the loan/custody event. All users share the same ARTIFACT_ID, LOAN_ID and event dates so that the loan register, gallery plan and condition evidence refer to the same object.

1.2 Three alternative conceptual designs

Alternative	Core representation	Strength	Failure/counterexample
A. Dated relationship model	LENDER —< LOAN_EVENT >— ARTIFACT, with start/end dates on the relationship; placement and reports are separate.	Represents a dated custody relationship without creating an independent loan object.	A loan containing many artifacts needs repeated relationship rows; a single loan-level agreement number and status become awkward to manage.
B. Independent LOAN occurrence (preferred)	LENDER 1—N LOAN; LOAN 1—N LOAN_OBJECT; ARTIFACT 1—N LOAN_OBJECT over time.	A loan is an independently identified operational event; each artifact line has its own dates/evidence link.	Requires LOAN_OBJECT rows and constraints to prevent overlapping active custody for the same artifact.
C. Shared PERSON identity with role subtypes	PERSON is the common identity; LENDER and EMPLOYEE roles refer to PERSON; EMPLOYEE specializes into CURATOR and REGISTRAR.	Avoids duplicate people when a curator lends a privately owned artifact; contact identity is consistent.	A design with separate unrelated LENDER and EMPLOYEE tables can create duplicate people and inconsistent contact details.

The alternatives differ structurally, not merely by renaming LOAN. Alternative A makes the loan a dated association. Alternative B gives each loan an independent identity and supports a multi-artifact loan. Alternative C changes the identity model by generalising overlapping roles into PERSON. The preferred complete model combines B and C.

1.3 Custody periods and condition evidence

Each LOAN has loan-level agreement dates, while LOAN_OBJECT records the actual custody interval for each artifact (received_at and due_at/returned_at). A CONDITION_REPORT belongs to one ARTIFACT and records the observed condition, inspection date and author. It may optionally reference the relevant LOAN_OBJECT. This allows reports before dispatch, at receipt, during display and at return to remain attached to the physical object even when its lender or ownership changes.

1.4 Ownership of LOAN_OBJECT evidence

LOAN_OBJECT is identified by the loan and artifact pair (or by a separate LoanObjectID). It is dependent on both LOAN and ARTIFACT. Evidence about the custody of a particular artifact is linked to that LOAN_OBJECT, not stored only under the lender. This prevents reports for two different objects from being mixed merely because they came from the same lender.

Concrete failure resolved: if lender A lends artifact X and later lends artifact Y, a report filed only under lender A cannot reliably establish which object was inspected. Linking the report to ARTIFACT_ID and optionally LOAN_OBJECT_ID resolves the ambiguity.

TASK 2 – Entities, Attributes and Keys

Entity	Primary key	Important attributes / key reasoning
PERSON	PersonID	GivenName, MiddleName, FamilyName; one identity even if the person has several roles.
ORGANIZATION	OrganizationID	LegalName, address; separate identity from a natural person.
EMPLOYEE	PersonID (also FK)	EmployeeNo (unique), HireDate, Role; subtype of PERSON.
CURATOR	PersonID (FK)	SpecialistArea; employee subtype.
REGISTRAR	PersonID (FK)	PermissionLevel; employee subtype.
LENDER	LenderID	PersonID or OrganizationID reference; lender may be an individual or organisation.
ARTIFACT	ArtifactID	AccessionNo (unique), Title, CollectionTypeID, Description.
LOAN	LoanID	AgreementNo (unique), LenderID, StartDate, EndDate, Status.
LOAN_OBJECT	LoanObjectID	LoanID, ArtifactID, ReceivedAt, DueAt, ReturnedAt, agreed loan-line value.
EXHIBITION	ExhibitionID	Name, StartDate, EndDate, Venue.
DISPLAY_SLOT	ExhibitionID + SlotNo	Weak entity; SlotNo is unique only within its exhibition.
CONDITION_REPORT	ReportID	ArtifactID, optional LoanObjectID, ReportDate, ConditionText, CreatedBy.
ARTIFACT_OWNER	ArtifactID + OwnerID + ValidFrom	Owner share and ownership validity period; history of legal ownership.
PERSON_PHONE	PersonID + PhoneNumber	One phone number per row; supports multiple numbers.

2.1 Attribute types

- Contact information is multivalued when a person or organisation can have several phone numbers or email addresses. Store each value in a separate child relation.
- A person's name is composite: given name, optional middle name and family name.
- CollectionType is a coded concept. If it has its own description, retention rules or reporting relationships, represent it as COLLECTION_TYPE with CollectionTypeID.
- Loan start, due and return dates are stored source facts. Loan duration, current overdue status and the number of reports are derived/calculated values.
- A report's condition text and inspection date are stored facts, not derived attributes.

2.2 Candidate and owner-scoped keys

Stable surrogate primary keys such as ArtifactID, LoanID and ExhibitionID are preferred because names, descriptions and contact details can change. AccessionNo and AgreementNo are business candidate keys when the museum guarantees their uniqueness. An exhibition name cannot identify a display slot: one exhibition contains multiple slots, and different exhibitions may use the same slot number. DISPLAY_SLOT therefore uses the owner-scoped key (ExhibitionID, SlotNo).

Concurrent registration should use database-generated AUTO_INCREMENT or UUID identifiers, UNIQUE constraints on business keys and transactions. The database's unique index is the final authority; application-side 'check then insert' alone is unsafe because two users can check simultaneously.

TASK 3 – Relationships and Structural Constraints

3.1 Binary and ternary relationships

The main relationships are binary: LENDER gives LOAN; LOAN contains LOAN_OBJECT; ARTIFACT appears on LOAN_OBJECT; EXHIBITION owns DISPLAY_SLOT; ARTIFACT is placed in DISPLAY_SLOT; and ARTIFACT has CONDITION_REPORT. A possible ternary fact such as (LENDER, CURATOR, EXHIBITION) should be used only if the museum records one event whose meaning depends simultaneously on all three participants, for example a curator arranging a lender's artifact for a named exhibition. In the preferred model, this is normally decomposed into LOAN, CURATOR assignment and PLACEMENT because each has its own lifecycle and attributes.

3.2 Participation and cardinality

Relationship	Minimum / maximum participation	Reason
LENDER — LOAN	LENDER 0..N; LOAN exactly 1	A lender may never lend; every loan must identify one lender.
LOAN — LOAN_OBJECT	LOAN 1..N (after confirmation); LOAN_OBJECT exactly 1	A confirmed loan must contain at least one artifact; each line belongs to one loan.
ARTIFACT — LOAN_OBJECT	ARTIFACT 0..N over history; LOAN_OBJECT exactly 1	An artifact may have no loan yet or successive loans; each line identifies one artifact.
ARTIFACT — CONDITION_REPORT	ARTIFACT 0..N; CONDITION_REPORT exactly 1	A report must concern one physical artifact; an artifact may not yet have a report.
EXHIBITION — DISPLAY_SLOT	EXHIBITION 0..N; DISPLAY_SLOT exactly 1	A slot cannot exist independently of its exhibition.
EXHIBITION — CURATOR	EXHIBITION 1..N planned staff (business rule); CURATOR 0..N	A curator may work on multiple exhibitions; an exhibition needs assigned planning responsibility.

3.3 Where facts belong

Agreement-level dates and status belong on LOAN. Artifact-specific receipt, due and return dates, and loan-line valuation belong on LOAN_OBJECT. Placement dates and position belong on the ARTIFACT–DISPLAY_SLOT placement association. Inspection date, condition and inspector belong on CONDITION_REPORT. Ownership percentage and validity dates belong on ARTIFACT_OWNER.

3.4 Ternary projection counterexample

Consider a ternary relation R(Lender, Curator, Exhibition) with tuples (L1,C1,E1) and (L1,C2,E2). Its binary projections are LC={(L1,C1),(L1,C2)}, LE={(L1,E1),(L1,E2)} and CE={(C1,E1),(C2,E2)}. Joining these projections can create spurious tuples (L1,C1,E2) and (L1,C2,E1), which were not in R. Therefore the three binary projections do not always reconstruct the ternary fact. Use a ternary relation only when that exact combined fact is required; otherwise model independent operational events separately.

Missing LOAN_OBJECT evidence is significant: a LOAN header without artifact lines cannot show what was actually received. A confirmed loan should therefore require at least one line, enforced by a transaction or application workflow because a simple row CHECK cannot count child rows.

TASK 4 – Chen ER Diagram and Attribute Catalogue

4.1 Textual Chen-style ER representation

```
PERSON (PersonID PK) ──< EMPLOYEE (PersonID PK/FK) ── ISA ── CURATOR
                                                    └──────────── REGISTRAR

PERSON / ORGANIZATION ── LENDER ──< LOAN ──< LOAN_OBJECT >── ARTIFACT

ARTIFACT ──< CONDITION_REPORT

EXHIBITION ── identifies ──< DISPLAY_SLOT (weak; partial key SlotNo)

ARTIFACT ──< PLACEMENT >── DISPLAY_SLOT

ARTIFACT ──< ARTIFACT_OWNER >── PERSON / ORGANIZATION

PERSON ──< PERSON_PHONE
```

Legend: PK = primary key; FK = foreign key; ISA = specialization; < means one-to-many; DISPLAY_SLOT is weak and identified by its EXHIBITION owner plus SlotNo. Repeated references to PERSON, ARTIFACT or EXHIBITION in a drawing are aliases of the same entity, not duplicate entities.

4.2 Complete core attribute catalogue

Entity	Attributes
PERSON	PersonID (PK), GivenName, MiddleName, FamilyName, Email (if single-valued), Address
PERSON_PHONE	PersonID (PK/FK), PhoneNumber (PK), PhoneType
ORGANIZATION	OrganizationID (PK), LegalName, Address, Email
EMPLOYEE	PersonID (PK/FK), EmployeeNo (UNIQUE), HireDate, Role
CURATOR	PersonID (PK/FK), SpecialistArea
REGISTRAR	PersonID (PK/FK), PermissionLevel
LENDER	LenderID (PK), PersonID (nullable FK), OrganizationID (nullable FK), CHECK exactly one is present
ARTIFACT	ArtifactID (PK), AccessionNo (UNIQUE), Title, Description, CollectionTypeID (FK), CurrentStatus
LOAN	LoanID (PK), AgreementNo (UNIQUE), LenderID (FK), StartDate, EndDate, Status, CreatedBy (FK)
LOAN_OBJECT	LoanObjectID (PK), LoanID (FK), ArtifactID (FK), ReceivedAt, DueAt, ReturnedAt, LoanLineValue
EXHIBITION	ExhibitionID (PK), Name, StartDate, EndDate, Venue
DISPLAY_SLOT	ExhibitionID (PK/FK), SlotNo (PK), Gallery, PositionDescription
PLACEMENT	PlacementID (PK), ArtifactID (FK), ExhibitionID + SlotNo (composite FK), FromDate, ToDate
CONDITION_REPORT	ReportID (PK), ArtifactID (FK), LoanObjectID (nullable FK), ReportDate, ConditionText, CreatedBy (FK)
COLLECTION_TYPE	CollectionTypeID (PK), TypeName (UNIQUE), Description
ARTIFACT_OWNER	ArtifactID (PK/FK), OwnerID (PK), ValidFrom (PK), ValidTo, SharePercent

A placement must reference the composite key (ExhibitionID, SlotNo), not SlotNo alone. This prevents an artifact from being assigned to a slot number belonging to a different exhibition.

TASK 5 – EER Model: Generalization, Specialization and Category

PERSON is the supertype for people who may act as lenders or employees. EMPLOYEE is a subtype of PERSON. EMPLOYEE is further specialized into CURATOR and REGISTRAR. A curator who lends a privately owned object is one PERSON with both employee/curator and lender roles; the design does not create two separate people.

The CURATOR and REGISTRAR subtypes are generally disjoint if the museum's employee Role field permits exactly one of those roles at a time. If staff can perform both roles concurrently, the specialization must be overlapping and Role must be replaced by a many-to-many EMPLOYEE_ROLE relation. The specialization is partial if some employees have roles other than curator or registrar; it is total only if every employee must belong to one of the listed subtypes.

For the stated role-discriminator design, Role='CURATOR' creates a CURATOR subtype row and Role='REGISTRAR' creates a REGISTRAR row. The subtype row must have the same PersonID as EMPLOYEE. A trigger or controlled transaction is needed to keep the discriminator and subtype rows consistent.

A notification recipient may be either a PERSON or an ORGANIZATION. Model this as a union/category CONTACT_PARTY over PERSON and ORGANIZATION, or use a notification-recipient table with mutually exclusive PersonID and OrganizationID references and a CHECK that exactly one is non-null.

One wide table containing all lender, curator and registrar fields is inferior because it creates many NULL columns, duplicates shared identity/contact data, permits contradictory role combinations and makes a person's multiple roles difficult to represent. The subtype approach stores common attributes once and role-specific attributes only where applicable.

```
PERSON ── ISA (partial) ── EMPLOYEE ── ISA (disjoint, subject to policy) ── CURATOR
                                                                          └──────────── REGISTRAR

CONTACT_PARTY = PERSON ∪ ORGANIZATION
```

TASK 6 – Weak Entities and Complex Attributes

DISPLAY_SLOT is a weak entity because its identity depends on its exhibition. Its owner is EXHIBITION and its partial key is SlotNo; the full key is (ExhibitionID, SlotNo). The identifying relationship is stronger than an ordinary foreign key: the exhibition identifier forms part of the slot's primary key.

Multiple telephone numbers must not be stored as comma-separated text. PERSON_PHONE(PersonID, PhoneNumber, PhoneType) stores one number per row. This supports validation, searching and updates without parsing strings.

Names should be decomposed into GivenName, MiddleName and FamilyName when the application needs to sort, search or display them independently. Avoid assuming every person's name follows the same cultural format; retain a DisplayName if the source name must be preserved.

Derived summaries include loan duration (from dates), overdue status (current date compared with DueAt when ReturnedAt is null), total artifacts in a loan, number of condition reports and current owner-share total. Store the source dates and facts; calculate summaries in queries or views so they do not become stale.

CollectionType should be a separate entity when it has its own code, label, description or reporting use. Other codes such as LoanStatus or PhoneType can remain constrained lookup values if they have no independent attributes or relationships.

TASK 7 – Mapping the EER Model to Relational Schema

Strong entities map to relations with their own primary keys. Subtypes map to separate relations whose primary key is also a foreign key to the supertype. The weak DISPLAY_SLOT maps to a relation whose primary key contains the owner's key and the partial key. A 1:N relationship is mapped by placing the foreign key on the N-side. M:N relationships such as ARTIFACT–EXHIBITION placement use an associative relation. Multivalued phone numbers map to PERSON_PHONE.

7.1 Relational schema

```
PERSON(PersonID PK, GivenName, MiddleName, FamilyName, DisplayName, Email, Address)
PERSON_PHONE(PersonID PK/FK->PERSON, PhoneNumber PK, PhoneType)
ORGANIZATION(OrganizationID PK, LegalName, Address, Email)
EMPLOYEE(PersonID PK/FK->PERSON, EmployeeNo UNIQUE, HireDate, Role)
CURATOR(PersonID PK/FK->EMPLOYEE, SpecialistArea)
```

```
REGISTRAR(PersonID PK/FK->EMPLOYEE, PermissionLevel)
LENDER(LenderID PK, PersonID FK->PERSON NULL, OrganizationID FK->ORGANIZATION NULL)
COLLECTION_TYPE(CollectionTypeID PK, TypeName UNIQUE, Description)
ARTIFACT(ArtifactID PK, AccessionNo UNIQUE, Title, Description, CollectionTypeID FK)
LOAN(LoanID PK, AgreementNo UNIQUE, LenderID FK, StartDate, EndDate, Status, CreatedBy FK->EMPLOYEE)
LOAN_OBJECT(LoanObjectID PK, LoanID FK, ArtifactID FK, ReceivedAt, DueAt, ReturnedAt, LoanLineValue)
EXHIBITION(ExhibitionID PK, Name, StartDate, EndDate, Venue)
DISPLAY_SLOT(ExhibitionID PK/FK, SlotNo PK, Gallery, PositionDescription)
PLACEMENT(PlacementID PK, ArtifactID FK, ExhibitionID, SlotNo, FromDate, ToDate,
          FK(ExhibitionID,SlotNo)->DISPLAY_SLOT)
CONDITION_REPORT(ReportID PK, ArtifactID FK, LoanObjectID FK NULL, ReportDate, ConditionText, CreatedBy
FK->EMPLOYEE)
ARTIFACT_OWNER(ArtifactID PK/FK, OwnerID PK, ValidFrom PK, ValidTo, SharePercent)
```

For overlapping PERSON roles, separate subtype relations are appropriate because they preserve one person identity and allow multiple role memberships. Historical LOAN and LOAN_OBJECT rows should not be physically deleted when a loan closes. Use ReturnedAt and Status to close custody; use RESTRICT/soft deletion and audit records to preserve evidence. A lender or artifact referenced by historical loans should be deactivated rather than deleted.

TASK 8 – MySQL Implementation and Reproducible Instances

8.1 Core DDL

```
CREATE DATABASE meridian_museum;
USE meridian_museum;

CREATE TABLE Person (
  PersonID BIGINT PRIMARY KEY AUTO_INCREMENT,
  GivenName VARCHAR(80) NOT NULL,
  MiddleName VARCHAR(80),
  FamilyName VARCHAR(80),
  DisplayName VARCHAR(200)
);

CREATE TABLE Organization (
  OrganizationID BIGINT PRIMARY KEY AUTO_INCREMENT,
  LegalName VARCHAR(200) NOT NULL UNIQUE
);

CREATE TABLE Employee (
  PersonID BIGINT PRIMARY KEY,
  EmployeeNo VARCHAR(30) NOT NULL UNIQUE,
  HireDate DATE NOT NULL,
  Role VARCHAR(20) NOT NULL,
  CONSTRAINT fk_emp_person FOREIGN KEY (PersonID)
    REFERENCES Person(PersonID),
  CONSTRAINT chk_employee_role
    CHECK (Role IN ('CURATOR','REGISTRAR','OTHER'))
);

CREATE TABLE Curator (
  PersonID BIGINT PRIMARY KEY,
  SpecialistArea VARCHAR(120),
  FOREIGN KEY (PersonID) REFERENCES Employee(PersonID)
);

CREATE TABLE Registrar (
```

```
  PersonID BIGINT PRIMARY KEY,
  PermissionLevel VARCHAR(40),
  FOREIGN KEY (PersonID) REFERENCES Employee(PersonID)
);

CREATE TABLE Lender (
  LenderID BIGINT PRIMARY KEY AUTO_INCREMENT,
  PersonID BIGINT NULL,
  OrganizationID BIGINT NULL,
  CHECK ((PersonID IS NOT NULL AND OrganizationID IS NULL)
      OR (PersonID IS NULL AND OrganizationID IS NOT NULL)),
  FOREIGN KEY (PersonID) REFERENCES Person(PersonID),
  FOREIGN KEY (OrganizationID) REFERENCES Organization(OrganizationID)
);

CREATE TABLE Collection_Type (
  CollectionTypeID INT PRIMARY KEY AUTO_INCREMENT,
  TypeName VARCHAR(80) NOT NULL UNIQUE,
  Description VARCHAR(255)
);

CREATE TABLE Artifact (
  ArtifactID BIGINT PRIMARY KEY AUTO_INCREMENT,
  AccessionNo VARCHAR(40) NOT NULL UNIQUE,
  Title VARCHAR(200) NOT NULL,
  Description TEXT,
  CollectionTypeID INT NOT NULL,
  FOREIGN KEY (CollectionTypeID)
    REFERENCES Collection_Type(CollectionTypeID)
);

CREATE TABLE Loan (
  LoanID BIGINT PRIMARY KEY AUTO_INCREMENT,
  AgreementNo VARCHAR(40) NOT NULL UNIQUE,
  LenderID BIGINT NOT NULL,
  StartDate DATE NOT NULL,
  EndDate DATE NOT NULL,
  Status VARCHAR(20) NOT NULL,
  CreatedBy BIGINT NOT NULL,
  CHECK (EndDate >= StartDate),
  CHECK (Status IN ('DRAFT','OPEN','CLOSED','CANCELLED')),
  FOREIGN KEY (LenderID) REFERENCES Lender(LenderID),
  FOREIGN KEY (CreatedBy) REFERENCES Employee(PersonID)
);

CREATE TABLE Loan_Object (
  LoanObjectID BIGINT PRIMARY KEY AUTO_INCREMENT,
  LoanID BIGINT NOT NULL,
  ArtifactID BIGINT NOT NULL,
  ReceivedAt DATE NOT NULL,
  DueAt DATE NOT NULL,
  ReturnedAt DATE NULL,
  LoanLineValue DECIMAL(12,2),
  UNIQUE (LoanID, ArtifactID, ReceivedAt),
  CHECK (DueAt >= ReceivedAt),
  CHECK (ReturnedAt IS NULL OR ReturnedAt >= ReceivedAt),
  FOREIGN KEY (LoanID) REFERENCES Loan(LoanID),
  FOREIGN KEY (ArtifactID) REFERENCES Artifact(ArtifactID)
);

CREATE TABLE Exhibition (
  ExhibitionID BIGINT PRIMARY KEY AUTO_INCREMENT,
  Name VARCHAR(160) NOT NULL,
  StartDate DATE NOT NULL,
```

```
  EndDate DATE NOT NULL,
  Venue VARCHAR(160),
  CHECK (EndDate >= StartDate)
);

CREATE TABLE Display_Slot (
  ExhibitionID BIGINT NOT NULL,
  SlotNo INT NOT NULL,
  Gallery VARCHAR(80),
  PositionDescription VARCHAR(160),
  PRIMARY KEY (ExhibitionID, SlotNo),
  FOREIGN KEY (ExhibitionID) REFERENCES Exhibition(ExhibitionID)
);

CREATE TABLE Placement (
  PlacementID BIGINT PRIMARY KEY AUTO_INCREMENT,
  ArtifactID BIGINT NOT NULL,
  ExhibitionID BIGINT NOT NULL,
  SlotNo INT NOT NULL,
  FromDate DATE NOT NULL,
  ToDate DATE,
  CHECK (ToDate IS NULL OR ToDate >= FromDate),
  FOREIGN KEY (ArtifactID) REFERENCES Artifact(ArtifactID),
  FOREIGN KEY (ExhibitionID, SlotNo)
    REFERENCES Display_Slot(ExhibitionID, SlotNo)
);

CREATE TABLE Condition_Report (
  ReportID BIGINT PRIMARY KEY AUTO_INCREMENT,
  ArtifactID BIGINT NOT NULL,
  LoanObjectID BIGINT NULL,
  ReportDate DATETIME NOT NULL,
  ConditionText TEXT NOT NULL,
  CreatedBy BIGINT NOT NULL,
  FOREIGN KEY (ArtifactID) REFERENCES Artifact(ArtifactID),
  FOREIGN KEY (LoanObjectID) REFERENCES Loan_Object(LoanObjectID),
  FOREIGN KEY (CreatedBy) REFERENCES Employee(PersonID)
);

CREATE TABLE Person_Phone (
  PersonID BIGINT NOT NULL,
  PhoneNumber VARCHAR(25) NOT NULL,
  PhoneType VARCHAR(20),
  PRIMARY KEY (PersonID, PhoneNumber),
  FOREIGN KEY (PersonID) REFERENCES Person(PersonID)
);
```

8.2 Row-level and cross-row validation

The CHECK constraints validate facts within one row, such as EndDate >= StartDate and DueAt >= ReceivedAt. Rules involving other rows or tables need transactions, triggers or application logic: every confirmed loan must contain at least one LOAN_OBJECT; one artifact must not have two overlapping active custody periods; ownership shares must total 100% for each effective date; and a condition report's LoanObjectID must refer to the same ArtifactID as the report.

8.3 Reproducible population strategy

Use a fixed random seed of 4211 in the data-generation program. Generate parent records first, then retain the generated primary keys in arrays/lists and use those keys when creating dependent rows. For example, create 100, 1,000, 10,000 and 100,000 lenders, then generate proportional loans, loan lines, artifacts and condition reports. Do not assume generated IDs are consecutive if rows may be skipped; select actual keys from the parent table.

```
-- Example reproducible SQL checks after loading each dataset
SELECT COUNT(*) AS lender_count FROM Lender;
SELECT COUNT(*) AS loan_count FROM Loan;
SELECT COUNT(*) AS loan_object_count FROM Loan_Object;
SELECT COUNT(*) AS report_count FROM Condition_Report;
```

Keep one schema unchanged across all four workload sizes. Only the number of inserted rows changes. Record the MySQL version, hardware, indexes, seed, row counts and execution method. Capture the actual error text from attempted duplicate accession numbers, invalid loan date ranges, invalid lender identity and invalid composite display-slot references. Do not invent error messages or claim that a write failed unless it was executed.

TASK 9 – Validation with Use-Case Queries

9.1 Eight operational SQL queries

1. Open overdue loans

```
SELECT l.LoanID, l.AgreementNo, lo.ArtifactID, lo.DueAt
FROM Loan l JOIN Loan_Object lo ON lo.LoanID=l.LoanID
WHERE lo.ReturnedAt IS NULL AND lo.DueAt < CURRENT_DATE()
  AND l.Status='OPEN';
```

2. Artifacts without condition evidence

```
SELECT a.ArtifactID, a.AccessionNo, a.Title
FROM Artifact a
LEFT JOIN Condition_Report cr ON cr.ArtifactID=a.ArtifactID
WHERE cr.ReportID IS NULL;
```

3. Artifacts currently on loan

```
SELECT DISTINCT a.ArtifactID, a.Title, l.AgreementNo, lo.DueAt
FROM Artifact a
JOIN Loan_Object lo ON lo.ArtifactID=a.ArtifactID
JOIN Loan l ON l.LoanID=lo.LoanID
WHERE lo.ReturnedAt IS NULL AND l.Status='OPEN';
```

4. Loan contents

```
SELECT l.AgreementNo, lo.ArtifactID, a.Title, lo.ReceivedAt,
       lo.DueAt, lo.ReturnedAt
FROM Loan l JOIN Loan_Object lo ON lo.LoanID=l.LoanID
JOIN Artifact a ON a.ArtifactID=lo.ArtifactID
ORDER BY l.LoanID, lo.ArtifactID;
```

5. Condition history for one artifact

```
SELECT cr.ReportDate, cr.ConditionText, cr.LoanObjectID
FROM Condition_Report cr
WHERE cr.ArtifactID = 1
ORDER BY cr.ReportDate;
```

6. Exhibition inventory by slot

```
SELECT e.Name, ds.SlotNo, a.AccessionNo, a.Title,
       p.FromDate, p.ToDate
FROM Placement p
JOIN Exhibition e ON e.ExhibitionID=p.ExhibitionID
JOIN Display_Slot ds ON ds.ExhibitionID=p.ExhibitionID
                    AND ds.SlotNo=p.SlotNo
JOIN Artifact a ON a.ArtifactID=p.ArtifactID
ORDER BY e.ExhibitionID, ds.SlotNo;
```

7. Lender loan summary

```
SELECT ln.LenderID, COUNT(DISTINCT l.LoanID) AS LoanCount,
       COUNT(lo.LoanObjectID) AS ArtifactLines
FROM Lender ln
LEFT JOIN Loan l ON l.LenderID=ln.LenderID
LEFT JOIN Loan_Object lo ON lo.LoanID=l.LoanID
GROUP BY ln.LenderID;
```

8. Ownership share validation

```
SELECT ArtifactID, SUM(SharePercent) AS TotalShare
FROM Artifact_Owner
WHERE ValidFrom <= CURRENT_DATE()
  AND (ValidTo IS NULL OR ValidTo >= CURRENT_DATE())
GROUP BY ArtifactID
HAVING SUM(SharePercent) <> 100.00;
```

9.2 Output and performance recording

For each query, record the actual result rows and result count at 100, 1,000, 10,000 and 100,000 lenders. An empty result is a valid result and should be shown as '0 rows'; it must not be omitted. Record explicit JOIN counts by counting the JOIN operations in the SQL statement. Measure elapsed time repeatedly under the same conditions and report the median, not a single best run.

Query	JOIN count	Rows returned	Median time (ms)
Q1 overdue open loans	1	Run in MySQL	Measure × repeated runs
Q2 artifacts without reports	1	Run in MySQL	Measure × repeated runs
Q3 artifacts currently on loan	2	Run in MySQL	Measure × repeated runs
Q4 loan contents	2	Run in MySQL	Measure × repeated runs
Q5 artifact condition history	0	Run in MySQL	Measure × repeated runs
Q6 exhibition inventory	3	Run in MySQL	Measure × repeated runs
Q7 lender summary	2	Run in MySQL	Measure × repeated runs
Q8 ownership share validation	0	Run in MySQL	Measure × repeated runs

9.3 Flat-file anomalies

A single flat LOAN file repeating lender name, address, artifact title, due date and condition report causes anomalies. Insertion anomaly: a lender cannot be stored until a loan/artifact row exists. Update anomaly: changing a lender's address requires updating every row and may leave inconsistent copies. Deletion anomaly: deleting the last loan row for a lender may accidentally delete the only copy of the lender's details. Separating PERSON/LENDER, LOAN, LOAN_OBJECT and CONDITION_REPORT removes these anomalies.

TASK 10 – File-Based, Hierarchical and Network Model Comparison

Model	Representation and access path	Limitations for Meridian
File-based	Separate lender, loan, artifact and report files linked by manually maintained identifiers.	Duplicate descriptions, weak referential integrity, update anomalies and difficult concurrent editing.
Hierarchical	A tree such as LENDER → LOAN → LOAN_OBJECT → ARTIFACT → CONDITION_REPORT; exhibition inventory is another tree.	A record naturally has one parent. Shared artifacts, multiple loans over time and shared ownership do not fit one simple tree without duplication or pointer work.
Network	Owner-member sets can connect LENDER to LOAN, LOAN to LOAN_OBJECT, ARTIFACT to reports and ARTIFACT to placements.	Supports multiple links better than a tree, but navigation is path-dependent and structural changes can affect application programs.
Relational DBMS	Relations connected through primary and foreign keys; SQL joins can follow different paths.	Requires indexes and query planning, but provides declarative queries, integrity constraints and transaction support.

A hierarchical path can efficiently list a lender's loans, but an artifact with successive loans and multiple condition reports may need repeated or cross-linked records. A network model can represent these links using owner-member sets, but

applications must navigate the correct sets. The relational model expresses the same questions with joins, such as overdue loans and artifacts without condition reports.

Three-schema architecture

- External level: operational views for registrars, curators and conservators. A curator view may show exhibition slots and artifact descriptions; a registrar view may show lender, loan dates and returns.
- Conceptual level: the shared logical schema containing PERSON, LENDER, LOAN, ARTIFACT, LOAN_OBJECT, EXHIBITION, DISPLAY_SLOT, PLACEMENT, CONDITION_REPORT and ownership history.
- Internal level: physical files, B-tree indexes, pages, buffers and access paths selected by the DBMS.

During a concurrent update, the transaction manager coordinates commit/rollback, the concurrency-control or lock manager prevents conflicting writes, the buffer manager handles cached pages, the storage manager writes records, and the recovery manager uses logs to recover committed work after failure. Unique constraints and foreign keys are checked as part of the write.

TASK 11 – Final Design Recommendation

The recommended design is the independent LOAN occurrence combined with shared PERSON identity and separate artifact-level custody evidence. It best represents one loan containing multiple artifacts, successive loans for the same artifact, a curator who also acts as a lender, and condition reports that belong to physical objects rather than merely to lenders.

The key traceability is: lender identity → LOAN; LOAN → LOAN_OBJECT → ARTIFACT; ARTIFACT → CONDITION_REPORT; EXHIBITION → DISPLAY_SLOT; ARTIFACT → PLACEMENT → DISPLAY_SLOT. This separates the legal relationship, custody period, exhibition placement and condition evidence.

The design's stable identifiers are PersonID, LenderID, ArtifactID, LoanID, ExhibitionID and ReportID. DISPLAY_SLOT uses (ExhibitionID, SlotNo). Business rules that need cross-row validation include non-overlapping active custody for an artifact, total ownership share of 100%, matching ArtifactID between a condition report and its referenced loan line, and at least one loan line for a confirmed loan.

At 100,000 lenders, indexes on LenderID, LoanID, ArtifactID, DueAt, ReturnedAt, ReportDate and the composite DISPLAY_SLOT key support common joins and filters. Query plans and timing must be measured; performance cannot be claimed from schema design alone. The design remains scalable in principle, but actual response times depend on hardware, indexes, data distribution and workload.

TASK 12 – Re-evaluate and Adapt the Design

A. Impact analysis

The original simplifying assumption that each artifact has one current lender/owner is no longer sufficient. Legal ownership can be shared and can change while a loan remains active. A single LenderID on ARTIFACT cannot represent multiple simultaneous owners, ownership percentages or ownership history. The affected concepts are ARTIFACT, LENDER/PERSON, the ownership association, loan permissions and temporal constraints. LOAN and LOAN_OBJECT still represent custody and must not be rewritten when ownership shares change.

The old identifier ArtifactID identifies the physical artifact but does not identify a particular ownership fact. Each ownership fact needs an owner, share and validity period. Loan custody is a separate dated event and can continue unchanged when the legal owners change.

B. Evaluation of the original design

The original exhibition representation remains suitable for exhibition placement: EXHIBITION identifies an event and DISPLAY_SLOT remains identified by (ExhibitionID, SlotNo). The changed requirement affects ownership, not the identity of an exhibition or its slots. Stable primary keys remain valid; the ownership cardinality and business constraints must change.

The dated relationship alternative can record changing ownership only if it is expanded into a temporal ownership association; otherwise it confuses ownership dates with loan dates. The independent LOAN occurrence alternative remains suitable for custody but needs a separate ownership-history relation. The shared PERSON identity alternative remains useful because an owner may also be an employee or curator.

C. Adaptation / redesign

Decision: MODIFY the preferred model. Add ARTIFACT_OWNER as an associative, temporal entity between ARTIFACT and an owner party (PERSON or ORGANIZATION). Each row stores SharePercent, ValidFrom and ValidTo. Use a key such as (ArtifactID, OwnerPartyID, ValidFrom), with non-overlapping validity intervals for the same artifact-owner pair. A separate OWNER_PARTY supertype/category can unify person and organisation owners.

```
ARTIFACT 1 ──< ARTIFACT_OWNER >── OWNER_PARTY
ARTIFACT_OWNER(ArtifactID, OwnerPartyID, ValidFrom, ValidTo, SharePercent)
```

The ownership association replaces the old single-current-lender ownership assumption. It does not replace LOAN or LOAN_OBJECT. To migrate, create an initial ARTIFACT_OWNER row for each existing artifact using the previously recorded owner and a suitable effective start date. Preserve the original loan and condition-report identifiers and dates. Do not infer undocumented historical ownership changes; mark unknown history explicitly and document the migration assumption.

D. Evidence-based defence

A migration validation query should detect artifacts whose active ownership shares do not total 100%, and overlapping ownership intervals. It should also verify that each existing LOAN_OBJECT and CONDITION_REPORT still points to the same artifact after migration.

```
-- Active ownership totals that are invalid
SELECT ArtifactID, SUM(SharePercent) AS ActiveShare
FROM Artifact_Owner
WHERE ValidFrom <= CURRENT_DATE()
  AND (ValidTo IS NULL OR ValidTo >= CURRENT_DATE())
GROUP BY ArtifactID
HAVING SUM(SharePercent) <> 100.00;

-- Custody evidence must still refer to the same artifact
SELECT cr.ReportID, cr.ArtifactID, lo.ArtifactID AS LoanLineArtifact
FROM Condition_Report cr
JOIN Loan_Object lo ON lo.LoanObjectID=cr.LoanObjectID
WHERE cr.ArtifactID <> lo.ArtifactID;
```

Generated data must now include one or more owner rows per artifact, ownership percentages that total 100% at each effective time, and ownership changes that can occur during an open loan. The number of ARTIFACT_OWNER rows will therefore be greater than or equal to the number of artifacts. Operational views can preserve their existing interface by continuing to expose the artifact's current owner or a formatted owner summary, while the underlying schema stores full history.

The additional association increases storage and introduces temporal integrity checks. Queries for current owners require date predicates and aggregation; indexes on (ArtifactID, ValidFrom, ValidTo) and (OwnerPartyID, ArtifactID) help access. The extra cost is justified because it prevents legal ownership history from being confused with custody. A loan can remain open across an ownership change, while the custody and condition evidence retain their original identifiers and dates.

Conclusion

The modified relational/EER design is the best fit for Meridian Museum. It gives each operational concept a clear identity: PERSON for human identity, LOAN for the borrowing agreement, LOAN_OBJECT for artifact-level custody, CONDITION_REPORT for physical evidence, DISPLAY_SLOT for exhibition-specific placement and ARTIFACT_OWNER for time-dependent legal ownership. The separation supports accurate history, referential integrity, reliable reporting and future growth.

Before submission, execute the DDL and eight queries in MySQL, generate the four required population sizes using seed 4211, paste actual outputs and error messages, and add screenshots of the ER/EER diagrams and query results. Replace any assumptions that differ from the lecturer's expected interpretation.